

PROJECT 3

# PROBLEM STATEMENT & STRATEGY OUTLINE

Producing Actionable Insights Using Data Digest Pipeline with NLP  
Query Chat Interface

AI-Powered Defence & Geopolitical Intelligence Platform

Prepared by Intern  
Divyansh Nagar  
September 2026

Purpose of this report: explain the problem, strategy, implemented intelligence pipeline, dashboard, and RAG-based GenAI chatbot in language that is useful to both technical and non-technical readers.

# 1. Executive Summary

Modern defence and strategic intelligence teams receive a very large amount of open-source information every day. The central problem is no longer simply finding information; it is deciding what matters, removing repeated reports, understanding the military and geographic context, estimating the level of concern, and turning the result into a usable briefing quickly.

Project 3 addresses this problem with an end-to-end platform that converts raw multi-source news and official information into structured intelligence. The system combines a 10-stage Data Digest Pipeline, a relational and vector data layer, an interactive tactical dashboard, and a Retrieval-Augmented Generation (RAG) conversational interface.

In simple terms: the pipeline acts as the processing factory, the database acts as the organized memory, the dashboard acts as the analyst's visual control room, and the GenAI chatbot acts as a natural-language investigation assistant.

## What the platform delivers

- **Multi-source ingestion:** continuously collects RSS/Atom feeds and configured sources such as PIB, MEA, MoD, defence portals, and global/regional news.
- **Noise reduction:** extracts clean article text, removes boilerplate, and semantically groups repeated wire stories.
- **Intelligence enrichment:** classifies articles, identifies military and geopolitical entities, resolves strategic locations, summarizes content, and analyzes threat and stance.
- **Threat visibility:** converts multiple signals into a normalized 0.0–1.0 Threat Score and four levels: CRITICAL, HIGH, MEDIUM, LOW.
- **Executive digests:** produces structured briefings with key takeaways and strategic implications.
- **Conversational retrieval:** allows analysts to ask natural-language questions and receive context-grounded answers with source traceability.
- **Command dashboard:** presents articles, threat information, geographic views/heatmaps, filtering, and live chat in one interface.

# 2. The Problem Being Solved

The project documents describe an intelligence environment characterized by information overload and noise saturation. Analysts may have to monitor dozens of portals and news sources while trying to detect meaningful developments under time pressure.

| Problem                    | What it means in practice                                                                                          | Why it matters                                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Multi-source fragmentation | Information is spread across official portals, RSS feeds, defence publications and news outlets.                   | Analysts must manually switch between sources and build the overall picture themselves. |
| Wire redundancy            | The same story can be republished with small headline or wording changes.                                          | Time is spent reviewing duplicates instead of new information.                          |
| Unstructured context       | Raw articles do not naturally expose units, weapons, leaders, borders or strategic geography as structured fields. | Important relationships and patterns are harder to search and compare.                  |
| Delayed threat assessment  | The significance of an event is often buried in long-form text.                                                    | Critical signals can be harder to spot quickly.                                         |
| Keyword-only retrieval     | Exact words do not always express the user's actual intelligence question.                                         | Situational questions require contextual/semantic retrieval.                            |

| Problem              | What it means in practice                                               | Why it matters                                           |
|----------------------|-------------------------------------------------------------------------|----------------------------------------------------------|
| Manual briefing work | Analysts must repeatedly summarize, organize and prepare daily digests. | Consumes analyst bandwidth and introduces inconsistency. |

### 3. Strategy: From Raw Information to Actionable Intelligence

The strategy is to separate the problem into modular stages. Each stage has a specific purpose and produces a cleaner, more useful representation for the next stage. This makes the system easier to test, operate, and improve.

| Layer                       | Plain-language purpose                                                           | Main technologies                                           |
|-----------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1. Ingestion & Extraction   | Bring information into the platform and retrieve the useful article text.        | feedparser, requests/urllib3, trafilatura, fallback parsers |
| 2. Cleaning & Deduplication | Remove junk and recognize stories that are effectively the same.                 | Regex/rules, sentence-transformers, cosine similarity       |
| 3. Cognitive NLP            | Understand topic, people/units/weapons, locations, summaries, threat and stance. | BART, spaCy, gazetteers, Gemini, rules/heuristics           |
| 4. Storage & Indexing       | Store structured records and make article meaning searchable.                    | SQLite, SQLAlchemy, ChromaDB                                |
| 5. Serving & Interaction    | Expose intelligence through APIs, dashboard and conversational chat.             | FastAPI, Pydantic, React 18, RAG engine                     |

Design principle: automation is used to reduce repetitive information handling, while source traceability and structured outputs keep the result inspectable by an analyst.

## 4. The 10-Stage Data Digest Intelligence Pipeline

The pipeline is the core processing engine. It transforms raw feed entries into structured intelligence records and digest-ready outputs. The project README describes the stages as modular and capable of running as an end-to-end orchestration or as standalone components.

| Stage                           | Technology                      | What it does                                                                                                            | Output / value                    |
|---------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>1. Ingestion</b>             | feedparser                      | Reads configured RSS/Atom feeds, source priorities, timestamps and identifiers.                                         | Raw feed entries                  |
| <b>2. Extraction</b>            | trafilatura + fallbacks         | Fetches article pages and extracts the main text while removing advertisements, navigation and tracking clutter.        | Clean article body                |
| <b>3. Cleaning</b>              | Regex + sanitizers              | Removes wire artifacts, newsletter boilerplate, unwanted Unicode and formatting noise; normalizes text.                 | Canonical cleaned text            |
| <b>4. Deduplication</b>         | sentence-transformers / MiniLM  | Creates semantic embeddings and uses cosine similarity to group articles that express essentially the same story.       | Canonical story clusters          |
| <b>5. Classification</b>        | BART zero-shot                  | Assigns one or more defence/geopolitical categories without requiring a separate trained classifier for every category. | 20 primary + 50+ sub-categories   |
| <b>6. NER &amp; Geo</b>         | spaCy + military/geo gazetteers | Finds military entities, decision-makers, weapons, vessels, aircraft, borders, bases and strategic locations.           | Structured entities + locations   |
| <b>7. Summarization</b>         | Gemini + TextRank fallback      | Creates concise executive summaries, key takeaways and strategic summaries.                                             | 3-bullet executive brief          |
| <b>8. Threat Analysis</b>       | Deterministic matrix + LLM      | Combines threat indicators into a 0.0–1.0 score and assigns CRITICAL/HIGH/MEDIUM/LOW.                                   | Threat score + implications       |
| <b>9. Sentiment/Stance</b>      | BART / VADER / rules            | Assesses reporting tone and geopolitical stance such as escalating or de-escalating.                                    | Tone/stance signals               |
| <b>10. Storage &amp; Digest</b> | SQLite + ChromaDB + Jinja2      | Stores structured records, embeds article chunks for retrieval, and generates digest files.                             | Database + vector index + digests |

### 4.1 Why the pipeline matters

- It prevents the dashboard and chatbot from having to work directly on messy raw feeds.
- It creates consistent metadata that can be filtered by date, category, region and threat level.
- It reduces repeated stories before analysts spend time reviewing them.
- It converts narrative text into searchable concepts such as military entities and locations.
- It produces machine-readable outputs that can feed both the dashboard and the RAG retrieval layer.

### 4.2 Important technical terms — explained simply

| Term                          | Easy explanation                                                                                  |
|-------------------------------|---------------------------------------------------------------------------------------------------|
| <b>RSS / Atom feed</b>        | A standardized way for a website to publish new items so software can collect them automatically. |
| <b>NLP</b>                    | Natural Language Processing: methods that allow software to analyze human-written text.           |
| <b>Embedding</b>              | A numerical representation of text meaning. Similar meanings tend to have nearby vectors.         |
| <b>Cosine similarity</b>      | A mathematical measure used here to estimate how similar two text vectors are.                    |
| <b>Semantic deduplication</b> | Finding duplicate stories based on meaning, not only exact matching words.                        |

| Term                     | Easy explanation                                                                                                               |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Zero-shot classification | Assigning text to categories using a model and category descriptions without training a new classifier for those exact labels. |
| NER                      | Named Entity Recognition: identifying names and typed entities such as people, organizations, equipment or locations.          |
| Gazetteer                | A controlled reference list used to improve recognition and disambiguation of known geographic or domain entities.             |
| LLM                      | Large Language Model: a generative model used here for summarization, strategic reasoning and conversational answers.          |
| Vector database          | A database optimized for searching numerical embeddings by semantic similarity; ChromaDB is used here.                         |
| Relational database      | A structured database storing records in tables with defined fields; SQLite is used here.                                      |
| RAG                      | Retrieval-Augmented Generation: retrieve relevant source information first, then ask the LLM to answer using that evidence.    |
| REST API                 | A web interface through which software components exchange data using HTTP endpoints.                                          |

## 5. Interactive Tactical Command Dashboard

The dashboard is the visual layer of the platform. Its role is not to replace the underlying intelligence processing; it makes the processed results easier to inspect, filter and investigate. The README specifies a React 18 dashboard with an intelligence feed, threat heatmaps, geographic maps, filtering matrices, article inspection, digest viewing and live RAG chat.

| Dashboard capability  | What the user sees                                                         | How the pipeline enables it                                                                 |
|-----------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Intelligence Feed     | Processed articles with structured metadata and threat information.        | Stages 1-10 create normalized article records in SQLite.                                    |
| Threat view / heatmap | A visual way to understand where higher-risk signals are concentrated.     | Threat scoring plus location extraction provide the underlying score and geographic fields. |
| Geographic map        | Articles/events can be connected to extracted strategic locations.         | NER + location extraction + gazetteers create geographic entities/coordinates.              |
| Filtering matrix      | Narrow the intelligence set by date, category, threat level and region.    | Classification, threat scoring and stored metadata become queryable filters.                |
| Article inspection    | Review the processed article and its derived intelligence fields.          | Extraction, cleaning, NER, summary, threat and stance stages enrich each record.            |
| Digest viewer         | Read generated daily/thematic intelligence briefs.                         | Digest generator compiles processed records into Markdown/JSON outputs.                     |
| Live chat             | Ask intelligence questions without manually constructing database queries. | FastAPI routes requests to the RAG query engine.                                            |

### 5.1 User journey through the dashboard

1. Information arrives from configured sources.
2. The pipeline cleans and enriches the articles.
3. Duplicate stories are clustered so the analyst sees a more useful canonical view.
4. Threat, topic, entity and location information become filters and visual signals.
5. The analyst opens an article, reviews the summary and strategic implications, and can move directly into a conversational question.
6. The chatbot retrieves supporting records and returns an answer with source references.

### 5.2 Why a dashboard is important for non-technical users

A database and API are powerful but require technical knowledge. The dashboard converts those underlying capabilities into visible controls and views. An analyst does not need to know SQL, vector mathematics or API syntax to filter intelligence, inspect a threat signal, or ask a question.

## 6. GenAI Chatbot: Retrieval-Augmented Intelligence

The chatbot is not designed as a general-purpose chatbot that simply generates an answer from its internal language-model knowledge. The project uses RAG so the system first retrieves relevant intelligence from its own processed archive and then constructs a constrained prompt for the LLM.

| RAG step          | Purpose                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------|
| 1. Query analysis | The system interprets the user's intent and can identify time, geography, category and threat constraints. |

| RAG step                    | Purpose                                                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 2. Hybrid retrieval         | ChromaDB finds semantically related article chunks, while SQLite applies structured filters such as dates, categories and threat tiers. |
| 3. Context grounding        | The retrieved evidence is assembled into a context package with identifiers.                                                            |
| 4. Constrained generation   | The LLM is instructed to answer from the supplied context rather than freely inventing supporting facts.                                |
| 5. Citation mapping         | The response is linked back to article title, source, date, URL and threat information so the analyst can trace the evidence.           |
| 6. Conversational follow-up | Dialogue history supports multi-turn investigation instead of forcing the analyst to restate every question.                            |

Example of how an analyst uses it

Question: “Summarize recent LAC border infrastructure developments.”

The intended flow is: identify the time/geographic intent → retrieve relevant article chunks from ChromaDB → use SQLite metadata to constrain the result → build a grounded context → ask Gemini for a concise synthesis → return the answer with source IDs/citation cards. This means the chatbot is an investigation interface over the platform's intelligence archive.

Why RAG is strategically useful

- **Grounding:** answers are based on retrieved project data rather than only model memory.
- **Traceability:** the user can inspect the supporting source information.
- **Natural language:** users can ask questions in ordinary language instead of constructing database queries.
- **Hybrid retrieval:** semantic meaning and structured filters are used together.
- **Offline fallback:** the documented system includes rule-based query support when LLM APIs are disconnected.

## 7. Technical Architecture in Plain English

The system is divided into five major layers. This separation keeps data collection, intelligence processing, storage, and user interaction independently manageable.

| Layer                  | Main components                                                                                 | Plain-language role                                                     |
|------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Ingestion & extraction | feedparser, requests/urllib3, trafilatura, fallback extractors                                  | Collect source entries and turn webpages into usable article text.      |
| Processing & NLP       | cleaner, deduplicator, BART, spaCy, gazetteers, summarizer, threat analyzer, sentiment analyzer | Turn text into structured intelligence.                                 |
| Storage & indexing     | SQLite, SQLAlchemy, ChromaDB, Jinja2                                                            | Keep records organized and make meaning-based retrieval fast.           |
| Backend serving        | FastAPI, Uvicorn, Pydantic v2                                                                   | Provide programmatic endpoints for articles, threats, chat and digests. |
| Frontend               | React 18, Vite, Lucide Icons, Vanilla CSS                                                       | Provide the interactive analyst-facing command dashboard.               |

## 8. Performance & Operational Targets

| Metric                        | Documented target / benchmark                   | How it is achieved                                                     |
|-------------------------------|-------------------------------------------------|------------------------------------------------------------------------|
| Deduplication                 | >85% duplicate reduction                        | Sentence-transformer embeddings + cosine similarity threshold of 0.82. |
| Extraction reliability        | >95% full-text yield                            | trafilatura with fallback extraction.                                  |
| Taxonomy                      | 20 primary + 50+ sub-categories                 | BART zero-shot multi-label scoring.                                    |
| Threat scoring                | 4-tier normalized scale                         | Deterministic keyword/weight matrix plus LLM strategic reasoning.      |
| Chat groundedness             | Zero unverified hallucinations (target)         | Strict RAG constraints and article-ID citation matching.               |
| Pipeline throughput           | Batch and incremental execution                 | Modular stages and offline fallback mode.                              |
| Documented ingestion capacity | 10,000+ articles/hour automated batch ingestion | Automated pipeline processing rather than manual review.               |

Important interpretation: these are the project's documented targets/benchmarks. They should be treated as engineering evaluation criteria and not as a guarantee of real-world intelligence accuracy.

## 9. Operational Value: Before vs. After

| Dimension          | Traditional/manual approach                         | Platform approach                                              |
|--------------------|-----------------------------------------------------|----------------------------------------------------------------|
| Data throughput    | ~20-50 articles/day/analyst (documented comparison) | 10,000+ articles/hour automated ingestion benchmark            |
| Duplicate handling | Manual reading across browser tabs                  | Vector-based semantic clustering                               |
| Threat visibility  | Qualitative/manual review                           | Normalized 0.0-1.0 score + four threat tiers                   |
| Entity mapping     | Manual highlighting and lookup                      | Automated NER + military/geographic gazetteers                 |
| Daily briefing     | 2-4 hours documented comparison                     | Automated synthesis documented as sub-second / under 5 seconds |
| Investigation      | Keyword search                                      | Conversational RAG with citations                              |



# 10. What Is Implemented and How the Pieces Connect

The supplied project documentation describes an implemented software structure rather than only a conceptual design. The repository contains dedicated modules for ingestion, NLP, processing, storage, digest generation, API serving, frontend components and RAG chat.

| Component             | Documented location                                        | Purpose                                                          |
|-----------------------|------------------------------------------------------------|------------------------------------------------------------------|
| Pipeline orchestrator | src/pipeline.py                                            | Runs the 10-stage processing workflow.                           |
| Feed reader           | src/ingestion/feed_reader.py                               | Reads configured RSS/Atom feeds.                                 |
| Article fetcher       | src/ingestion/article_fetcher.py                           | Extracts full article text.                                      |
| Cleaner               | src/processing/cleaner.py                                  | Sanitizes and normalizes text.                                   |
| Deduplicator          | src/processing/deduplicator.py                             | Performs vector cosine similarity deduplication.                 |
| Classifier            | src/nlp/classifier.py                                      | Performs zero-shot multi-label taxonomy classification.          |
| NER / location        | src/nlp/ner_extractor.py;<br>src/nlp/location_extractor.py | Finds military/geopolitical entities and strategic geography.    |
| Summarizer            | src/nlp/summarizer.py                                      | Creates executive summaries using Gemini or fallback extraction. |
| Threat analyzer       | src/nlp/threat_analyzer.py                                 | Computes threat score and strategic impact.                      |
| Sentiment analyzer    | src/nlp/sentiment_analyzer.py                              | Evaluates tone and geopolitical stance.                          |
| Database              | src/storage/database.py                                    | Defines structured storage models and indexing.                  |
| Digest generator      | src/digest/generator.py                                    | Compiles daily/thematic intelligence briefings.                  |
| RAG engine            | src/chat/query_engine.py                                   | Runs retrieval, grounding and citation construction.             |
| FastAPI backend       | src/api/main.py                                            | Exposes article, threat, chat and digest endpoints.              |
| React frontend        | frontend/src/                                              | Provides the analyst dashboard, chat and visual components.      |

# 11. Strategy for Non-Technical Stakeholders

What problem does it solve? — It reduces the amount of manual reading and sorting required before an analyst can understand what is happening.

What is the pipeline? — A sequence of automated processing steps that turns raw news into cleaned, classified, summarized and risk-tagged intelligence.

What is the dashboard? — A visual command surface where analysts can inspect the intelligence feed, threats, geography and filters.

What is the GenAI chatbot? — A question-and-answer interface that searches the project's intelligence archive before generating a response.

Why citations? — They allow the user to trace an answer back to the underlying article information.

Why two databases? — SQLite is suited to structured records and filters; ChromaDB is suited to meaning-based similarity search.

What does threat scoring do? — It provides a consistent numerical and tiered way to prioritize signals for review; it does not replace human judgment.

## 12. Strategic Outcomes

- Faster movement from raw information to an organized intelligence picture.
- Reduced analyst time spent on duplicate stories and repetitive briefing preparation.
- More consistent tagging of defence, border, maritime, cyber and other intelligence categories.
- Better visibility of military/geopolitical entities and strategic locations.
- Prioritized review through a normalized threat score and threat tiers.
- Natural-language access to the intelligence archive through RAG.
- Improved traceability because conversational answers can carry source references.
- A foundation that can scale toward richer data sources and more operational deployment modes.

## 13. Future Strategy / Scale-Out

| Future enhancement              | Strategic purpose                                                                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Air-gapped local LLM deployment | Use local quantised models such as LLaMA-3-8B-Instruct or Mistral-7B with vLLM/Ollama to reduce dependency on external APIs and support offline environments. |
| Multi-modal intelligence        | Extend beyond text into satellite imagery annotations, video/audio transcription and OCR.                                                                     |
| Knowledge graph                 | Represent relationships among states, commanders, organizations, weapon systems and procurement routes using a graph database such as Neo4j.                  |
| Automated alerts                | Trigger operational notifications when critical threat thresholds are exceeded; the documentation cites a >0.85 example threshold.                            |

## 14. Conclusion

Project 3 is structured as a complete intelligence workflow rather than a standalone scraper, NLP model, dashboard or chatbot. The strategic value comes from connecting these pieces: source ingestion supplies the information; the 10-stage pipeline turns it into structured intelligence; SQLite and ChromaDB preserve and index that intelligence; FastAPI exposes it; the React dashboard makes it understandable; and the RAG chatbot makes the archive conversational.

For technical stakeholders, the architecture provides modular processing stages, explicit technologies, data stores, APIs and a clear repository structure. For non-technical stakeholders, the same architecture can be understood as an automated path from information → filtering → understanding → prioritization → briefing → investigation.

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Date: September 2026